

codefest 7

1. a) $FLOPs = 2N^2 = 2 \times 512^2 = 524288$ Dense FLOPs

b) Dense Memory = $4 \times N^2 = 4 \times 512^2 = 1048576$ bytes

c) Sparse compute = $2N^2(1-s) = 524288(1-s)$

d) Value array = $4N^2(1-s)$, Column index array = $4N^2(1-s)$

Row pointer array = $4(N+1)$

$$\begin{aligned} \text{Sparse memory} &= 4N^2(1-s) + 4N^2(1-s) + 4(N+1) \\ &= 2097152 \times (1-s) + 4(513) = \\ &= 2097152(1-s) + 2052 \end{aligned}$$

2. $\text{Speedup} = \frac{\text{Dense FLOP}}{\text{Sparse FLOP}} = \frac{2N^2}{2N^2(1-s)} = \frac{1}{1-s} = 2 \Rightarrow 1 = 2 - 2s$
 $\Rightarrow s = \frac{1}{2}$

0.5 or 50% sparsity is required for x2 speedup

3. $1048576 = 2097152(1-s) + 2052 \Rightarrow 2097152(1-s) = 1046524$
 $\Rightarrow 1-s = \frac{1046524}{2097152} \Rightarrow 1-s = 0.499 \Rightarrow s = 0.501$

Above 50.1% sparsity sparse format uses less memory than dense storage

4. At $s=0.9$:

Sparse memory = $4(512)^2(1-0.9) + 4(512)^2(1-0.9) + (4 \times 513) = 211767.2$ bytes

Execution time of dense memory = $\frac{1048576 \text{ byte}}{320 \text{ Gb/s}} = 3.277 \mu s$

Execution time of sparse memory = $\frac{211767.2 \text{ bytes}}{320 \text{ Gb/s}} = 0.662 \mu s$

Speedup = $\frac{\text{Execution time of dense memory}}{\text{Execution time of sparse memory}} = \frac{3.277}{0.662} = 4.95$

At 90% sparsity on memory bandwidth limited system, the sparse MVM is 4.95 times faster than dense MVM